Unlicensed Substitution: Field Notes on How Language Models Replace the Standard Their Material Calls For

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Abstract

When a language model meets material it was not built around, it does not ask how that material wants to be read. It applies a standard it already carries, and it reports the result as warranted. I call this *unlicensed substitution*. This paper reports seven cases from three research projects that I designed and ran with several models: a critique of a manuscript grounded in Disability Critical Race Theory (DisCrit), a multi-model entry in the Kaggle "Squares in Squares" packing competition, and an origins-of-life simulation. In each case I record the standard the material called for, the default that replaced it, who caught the swap, and what corrected it. In this record, models caught one another's errors of fact and computation, while the human lead caught every error of frame. The model that detected others' errors committed four of the seven substitutions itself. In the one case tested in depth, explicit instruction in the framework's logic, given before critique, was sufficient to correct the misreading. The cases are selected, not sampled; I state the claims as propositions and close with four tests that could falsify them.

1 Proposition

Unlicensed substitution names a specific failure: a model replaces the standard its material calls for with a default standard, and nothing in the material licensed the swap. The working claim has three parts.

1. **Substitution occurs.** Models apply embedded standards, such as social-science methodology, numerical tolerance, confident completion, and softened or precautionary framing, where those standards do not fit.
2. **The model does not see it.** The output is internally consistent and presented as warranted. The swap is not visible from inside the response.
3. **Instruction can correct it.** Stating the framework's logic explicitly, before asking for critique, changed what the model could recognize as coherent in the one case tested in depth.

Part 3 has been tested on one framework. Whether it holds across frameworks, and whether the correction persists, is unknown.

The proposition grew out of an earlier observation: LLM critique of theoretically novel scholarship defaulted to the analytical apparatus the model carries rather than the framework's own logic (Case 1). The cases below test whether the same structure appears outside framework critique.

2 Setting and method

All three projects were designed and run by the author, who coordinated several models through a single "live thread" where work was collected, cross-checked, and decided. Other models received round prompts and returned reports. Two standards applied throughout: every number must come with a source or the code that reproduces it, and each report must separate what was verified, what a source states, and what is assumed.

For adversarial review, I used a *failure finder* role: the model starts from the assumption that every attached file and claim is wrong, succeeds only by producing a reproducible failure, and lists anything it could not break as surviving, with what it tried. Agreement does not count as a result. The full prompt is in the accompanying repository.

Project	Task	Models
Framework critique	Critique of a co-authored manuscript grounded in DisCrit [1] and Chicana feminist epistemology	Claude; two others in later checks
Squares in Squares	Pack n unit squares into the smallest square, $n = 1, \ldots, 1000$ [4, 2]	Claude, Grok, Gemini
Origins of life	Agent-based and reaction-network models of prebiotic chemistry	Claude

Table 1: Projects from which the cases are drawn.

3 Cases

Case 1: Mechanism-hunting in a framework that presupposes its conditions. DisCrit treats racism and ableism as foundational conditions; analysis attends to how people navigate and resist within a structure already presupposed as against them [1]. The model instead searched for racism as an isolable mechanism, found no discrete variable, and diagnosed that absence as a coherence problem in the manuscript. The author caught the substitution. Explicit instruction (*these are presupposed; do not look for them as mechanisms*) produced a full recalibration of the model's reading.

Case 2: Tolerance standing in for validity. The competition requires packings with no overlap; a competitor's repository reports that the official checker certifies non-overlap with interval arithmetic. One model's validator used a floating-point tolerance of 10^{-8}. Test inputs with overlaps of 5×10^{-9} and 10^{-12}, and a square 5×10^{-9} outside the container, were all accepted. A second model caught this by constructing the failing inputs. The correction was a reference validator using 50-digit decimal arithmetic that computes the side length from the square positions and reports VALID, CONTACT, or INVALID.

Case 3: "Best found" reported as "proven." One model's table listed $n = 11$, 17, and 19 as proven optimal. The Ellsworth catalogue [3] lists all three as unproven. A second model caught this by checking each row against the catalogue. The correction was a required proof-status column with a source for every row.

Case 4: One square measured as the whole packing. One model's optimization script computed the bounding box from a single square's width rather than the extent of the full packing.

A second model caught this by reading the code. A replacement optimizer (random-start L-BFGS followed by SLSQP with separating-line variables) reached the known optima for $n = 5$ and $n = 10$ from random starts and came within 0.25% of the record for $n = 11$.

Case 5: A reworded role that changed the role. Asked to remove wording likely to trigger automatic refusal, the coordinating model rewrote an adversarial reviewer role as "independent verification." The facts were unchanged, but the stance moved from *assume it is wrong and prove it* to *check it*, which permits the reviewer to go easy. The author caught the shift. The correction was the failure finder role described in Section 2.

Case 6: A precaution the rules did not call for. The competition rules allow external models and tools, including LLMs. The coordinating model nonetheless wrote the scope statement as a warning against manipulating the scorer, implying the project might be attempting to game it. The author caught this. The correction was a scope line stating the goal directly: valid, small packings that pass the official checker on their merits.

Case 7: Scripted results presented as emergence. The author's requirement was results arising from the dynamics, not the setup. Earlier runs replayed reactions written into the model. When a results chart was examined after the author's pushback, the model identified four setup artifacts: scheduled pulse jumps, a "functional" metric based on motifs that could occur by chance, an error test unable to detect a catastrophe, and a mislabeled parameter. Separately, source checks found two of four steps in a proposed energy-coupling mechanism unsupported as stated. One benchmark failure in the reaction engine remains open.

4 Findings

Case	Model	Default substituted	Caught by
1	Claude	Social-science mechanism-finding	Author
2	Grok	Floating-point tolerance	Model
3	Grok	Confident completion	Model
4	Gemini	Wrong measurement in code	Model
5	Claude	Softened stance	Author
6	Claude	Precautionary framing	Author
7	Claude	Scripted outcome read as emergent	Author, then model

Table 2: Substitutions, by model and by who detected them.

1. **Models caught each other's errors of fact and computation.** Cases 2–4 were found by one model testing another's code or claims against sources.
2. **The human lead caught every error of frame.** Cases 1 and 5–7 would pass a test suite: the outputs were internally consistent. What was wrong was the standard they were measured against, and no model in the loop flagged it.
3. **Detecting others' substitutions did not prevent one's own.** The model that found Cases 2–4 committed Cases 1 and 5–7.

These are patterns in a selected record of seven cases, not tested results.

5 Limitations

- **Sources.** The cases were reconstructed from session records and model-written session summaries. Not every transcript was re-read for this version.
- **Selection and absence.** Cases appear because someone noticed them. Following the author's reconstruction methodology, absence stays attached to the finding: a substitution not reported here was not found in the records searched, which does not mean it did not occur. Unnoticed substitution is the category the proposition predicts is largest.
- **Position of the drafting tool.** The model used to draft this text is a subject of four of the seven cases.
- **Agreement is weak evidence.** Models trained on overlapping material may converge because they share defaults, not because they independently recover the same finding. Multi-model agreement is treated here as weak evidence.

6 Tests that could falsify the proposition

1. **Across frameworks.** Repeat Case 1 with two or three other frameworks that presuppose their conditions. Record whether framework instruction suffices each time.
2. **Durability.** After instruction, introduce new material in the same framework without repeating the instruction. Record whether the model reverts.
3. **Self-detection.** Give a model its own output and the framework statement, and ask it to name the standard it applied. Record whether it can locate its own substitution.
4. **Frame errors under adversarial review.** Plant a frame substitution, rather than a computational error, in a round report and record whether the failure finder catches it.

AI assistance disclosure

The proposition, research design, coordination method, and identification of Cases 1 and 5–7 are the author's. Claude (Anthropic) drafted text from session records at the author's direction. Claude is also a subject of Cases 1 and 5–7. The author is responsible for all content.

Data and materials

Report, prompt, and source: `https://github.com/Choytcudz/fluffy-disco`. Licensed CC BY 4.0.

References

[1] S. A. Annamma, D. Connor, and B. Ferri. Dis/ability critical race studies (DisCrit): Theorizing at the intersections of race and dis/ability. *Race Ethnicity and Education*, 16(1):1–31, 2013. `https://doi.org/10.1080/13613324.2012.730511`

[2] E. Friedman. Packing unit squares in squares: A survey and new results. *Electronic Journal of Combinatorics*, Dynamic Survey DS7. `https://www.combinatorics.org/ojs/index.php/eljc/article/view/DS7`

[3] Ellsworth catalogue of squares in squares. https://kingbird.myphotos.cc/packing/squares_in_squares.html

[4] Kaggle. Squares in Squares (sponsored by Google LLC). https://www.kaggle.com/competitions/squares-in-squares